

TW3730TU – Numerical Analysis for PDE's

Lecture 3

Finite Difference Method in 1D: order of approximation, properties of 1D FD Laplacian

Finite-Difference Method in 1D

1D problem

We want to solve a 1D boundary value problem for the Poisson equation with Dirichlet BC's and different source functions:

$$-\frac{\partial^2 u}{\partial x^2} = f, \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

case 1: $f(x) = -1$

case 2: $f(x) = -x + x^2 - x^3 + x^4 - x^5 + x^6$

case 3: $f(x) = -\sin(\pi x)$

Analytical solutions

This is an ordinary linear second-order differential equation, which can be solved analytically:

case 1: $f(x) = -1$;

$$u = -\frac{1}{2}x + \frac{1}{2}x^2$$

case 2: $f(x) = -x + x^2 - x^3 + x^4 - x^5 + x^6$;

$$u = -\frac{1}{840}(15x^8 - 20x^7 + 28x^6 - 42x^5 + 70x^4 - 140x^3 + 89x)$$

case 3: $f(x) = -\sin(\pi x)$;

$$u = -\frac{1}{\pi^2} \sin(\pi x)$$

1D Grid

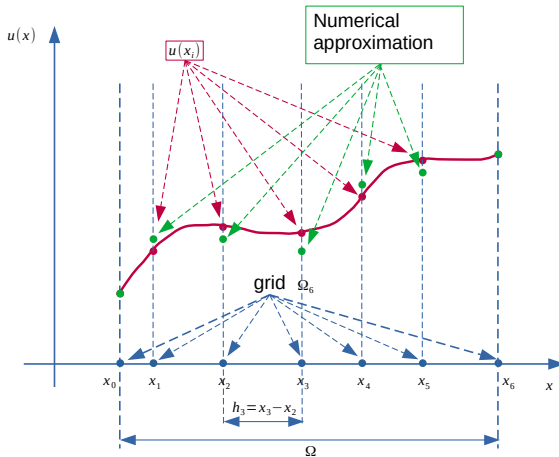
- A **grid** (mesh) on a segment $\Omega = [a, b]$ is any finite set of points of this segment. We denote as Ω_N the grid satisfying the following property:

$$a = x_0 < x_1 < x_2 < \cdots < x_{N-1} < x_N = b$$

These are the boundaries of the N sub-intervals dividing the domain Ω .

- Points x_i are called the **nodes** of the grid Ω_N
- We denote as f_i the **grid value** of the function $f(x)$ at the point $x_i \in \Omega_N$, i.e., $f_i = f(x_i)$ or $u_i = u(x_i)$.
- **Inner points/nodes:** x_i , $i = 1, \dots, N - 1$. **Boundary points/nodes** x_0 and x_N .

Idea of numerical solution



Equation at grid nodes

Let $u(x)$ be a sufficiently smooth function defined on Ω . Introducing Ω_h on Ω , $f_i = f(x_i)$ and $u_i = u(x_i)$ we write the PDE at x_i :

$$-u_i'' = f_i, \quad i = 1, 2, \dots, N-1,$$

and the two boundary conditions

$$u_0 = 0, \quad u_N = 0.$$

Here f_i , u_0 , and u_N are known. Unknowns are the values of u_i , $i = 1, \dots, N-1$; at the internal grid points x_i , $i = 1, \dots, N-1$.

We need a connection between the derivative u_i'' at x_i and the grid values u_i , $i = 0, \dots, N$. First we study the connection proposed by the Finite-Difference Method.

Deriving a Finite-Difference formula for $-u_i''$

Taylor series of $u(x)$ around x_i :

$$\begin{aligned}u(x) &= u(x_i) + u'(x_i)(x - x_i) + \frac{u''(x_i)}{2}(x - x_i)^2 + \frac{u'''(x_i)}{6}(x - x_i)^3 + \dots \\&= u_i + u'_i(x - x_i) + \frac{u''_i}{2}(x - x_i)^2 + \frac{u'''_i}{6}(x - x_i)^3 + \dots\end{aligned}$$

Take the nodes x_{i+1} and x_{i-1} :

$$\begin{aligned}u_{i+1} &= u_i + u'_i(x_{i+1} - x_i) + \frac{u''_i}{2}(x_{i+1} - x_i)^2 + \frac{u'''_i}{6}(x_{i+1} - x_i)^3 + \dots \\u_{i-1} &= u_i + u'_i(x_{i-1} - x_i) + \frac{u''_i}{2}(x_{i-1} - x_i)^2 + \frac{u'''_i}{6}(x_{i-1} - x_i)^3 + \dots\end{aligned}$$

Deriving a Finite-Difference formula for $-u_i''$

Rearrange:

$$\begin{aligned}-\frac{u_i''}{2}(x_{i+1} - x_i)^2 &= -u_{i+1} + u_i + u_i'(x_{i+1} - x_i) + \frac{u_i'''}{6}(x_{i+1} - x_i)^3 + \dots \\ -\frac{u_i''}{2}(x_{i-1} - x_i)^2 &= -u_{i-1} + u_i + u_i'(x_{i-1} - x_i) + \frac{u_i'''}{6}(x_{i-1} - x_i)^3 + \dots\end{aligned}$$

Divide:

$$\begin{aligned}-\frac{u_i''}{2} &= \frac{-u_{i+1} + u_i}{(x_{i+1} - x_i)^2} + u_i' \frac{x_{i+1} - x_i}{(x_{i+1} - x_i)^2} + \frac{u_i'''}{6} \frac{(x_{i+1} - x_i)^3}{(x_{i+1} - x_i)^2} + \dots \\ -\frac{u_i''}{2} &= \frac{-u_{i-1} + u_i}{(x_{i-1} - x_i)^2} + u_i' \frac{x_{i-1} - x_i}{(x_{i-1} - x_i)^2} + \frac{u_i'''}{6} \frac{(x_{i-1} - x_i)^3}{(x_{i-1} - x_i)^2} + \dots\end{aligned}$$

Deriving a Finite-Difference formula for $-u_i''$

Simplify:

$$-\frac{u_i''}{2} = \frac{-u_{i+1} + u_i}{(x_{i+1} - x_i)^2} + u_i' \frac{1}{x_{i+1} - x_i} + \frac{u_i'''}{6}(x_{i+1} - x_i) + \dots$$

$$-\frac{u_i''}{2} = \frac{-u_{i-1} + u_i}{(x_{i-1} - x_i)^2} + u_i' \frac{1}{x_{i-1} - x_i} + \frac{u_i'''}{6}(x_{i-1} - x_i) + \dots$$

Add:

$$\begin{aligned} -u_i'' &= \frac{-u_{i+1} + u_i}{(x_{i+1} - x_i)^2} + \frac{-u_{i-1} + u_i}{(x_{i-1} - x_i)^2} \\ &\quad + u_i' \frac{1}{x_{i+1} - x_i} + u_i' \frac{1}{x_{i-1} - x_i} \\ &\quad + \frac{u_i'''}{6}(x_{i+1} - x_i) + \frac{u_i'''}{6}(x_{i-1} - x_i) + \dots \end{aligned}$$

Finite-Difference approximation of $-u_i''$ on a uniform grid

Uniform grid: $x_{i+1} - x_i = h$, $i = 0, \dots, N$. Note that $x_{i-1} - x_i = -h$.

$$\begin{aligned} -u_i'' &= \frac{-u_{i+1} + u_i}{h^2} + \frac{-u_{i-1} + u_i}{h^2} \\ &\quad + u_i' \frac{1}{h} - u_i' \frac{1}{h} + \frac{u_i'''}{6} h - \frac{u_i'''}{6} h + \\ &\quad + \frac{u_i^{(4)}}{24} h^2 + \frac{u_i^{(4)}}{24} h^2 + \dots \end{aligned}$$

Hence,

$$-u_i'' = \frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} + \frac{u_i^{(4)}}{12} h^2 + \dots$$

Order of approximation

$$-u_i'' = \frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} + \mathcal{O}(h^2)$$

Here, $\mathcal{O}(h^2)$ is the **order of approximation**. It is the local approximation error:

$$\text{local approx. error} = \left| -u_i'' - \frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} \right| = \mathcal{O}(h^2)$$

This "central difference" formula is order- h^2 (second-order) approximation of $-u''$ at x_i , if the grid is uniform.

If we define the central-difference FD approximation of u_i'' as:

$$\mathcal{D}_{xx}^{(2)} u_i = \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2},$$

$$\text{then } \left| u_i'' - \mathcal{D}_{xx}^{(2)} u_i \right| = \mathcal{O}(h^2)$$

Finite-Difference approximations for u'_i

- The **left finite-difference derivative** (one-sided, backward)

$$\mathcal{D}_x^- u_i = \frac{u_i - u_{i-1}}{h}$$

- The **right finite-difference derivative** (one-sided, forward)

$$\mathcal{D}_x^+ u_i = \frac{u_{i+1} - u_i}{h}$$

- The **central finite-difference derivative** (two-sided)

$$\mathcal{D}_x^c u_i = \frac{u_{i+1} - u_{i-1}}{2h}$$

Derivation and order of $\mathcal{D}_x^- u_i$

Taylor series of $u(x)$ around x_i :

$$\begin{aligned} u(x) &= u(x_i) + u'(x_i)(x - x_i) + \frac{u''(x_i)}{2}(x - x_i)^2 + \frac{u'''(x_i)}{6}(x - x_i)^3 + \dots \\ &= u_i + u'_i(x - x_i) + \frac{u''_i}{2}(x - x_i)^2 + \frac{u'''_i}{6}(x - x_i)^3 + \dots \end{aligned}$$

Take the node x_{i-1} :

$$\begin{aligned} u_{i-1} &= u_i + u'_i(x_{i-1} - x_i) + \frac{u''_i}{2}(x_{i-1} - x_i)^2 + \frac{u'''_i}{6}(x_{i-1} - x_i)^3 + \dots \\ -u'_i(x_{i-1} - x_i) &= -u_{i-1} + u_i + \frac{u''_i}{2}(x_{i-1} - x_i)^2 + \frac{u'''_i}{6}(x_{i-1} - x_i)^3 + \dots \\ -u'_i &= \frac{-u_{i-1} + u_i}{x_{i-1} - x_i} + \frac{u''_i}{2}(x_{i-1} - x_i) + \frac{u'''_i}{6}(x_{i-1} - x_i)^2 + \dots \\ u'_i &= \frac{u_i - u_{i-1}}{h} + \mathcal{O}(h) \end{aligned}$$

Derivation and order of $\mathcal{D}_x^+ u_i$

Take the node x_{i+1} :

$$u_{i+1} = u_i + u'_i (x_{i+1} - x_i) + \frac{u''_i}{2} (x_{i+1} - x_i)^2 + \frac{u'''_i}{6} (x_{i+1} - x_i)^3 + \dots$$

$$-u'_i (x_{i+1} - x_i) = -u_{i+1} + u_i + \frac{u''_i}{2} (x_{i+1} - x_i)^2 + \frac{u'''_i}{6} (x_{i+1} - x_i)^3 + \dots$$

$$-u'_i = \frac{-u_{i+1} + u_i}{x_{i+1} - x_i} + \frac{u''_i}{2} (x_{i+1} - x_i) + \frac{u'''_i}{6} (x_{i+1} - x_i)^2 + \dots$$

$$u'_i = \frac{u_{i+1} - u_i}{h} + \mathcal{O}(h)$$

Derivation and order of $\mathcal{D}_x^c u_i$ on a uniform grid

Take the nodes x_{i-1} and x_{i+1} :

$$u_{i-1} = u_i - u_i' h + \frac{u_i''}{2} h^2 - \frac{u_i'''}{6} h^3 + \dots$$

$$u_{i+1} = u_i + u_i' h + \frac{u_i''}{2} h^2 + \frac{u_i'''}{6} h^3 + \dots$$

$$u_i' h = -u_{i-1} + u_i + \frac{u_i''}{2} h^2 - \frac{u_i'''}{6} h^3 + \dots$$

$$u_i' h = u_{i+1} - u_i - \frac{u_i''}{2} h^2 - \frac{u_i'''}{6} h^3 + \dots$$

$$2u_i' h = -u_{i-1} + u_{i+1} - \frac{u_i'''}{3} h^3 + \dots$$

$$u_i' = \frac{u_{i+1} - u_{i-1}}{2h} - \frac{u_i'''}{6} h^2 + \dots$$

$$u_i' = \frac{u_{i+1} - u_{i-1}}{2h} + \mathcal{O}(h^2)$$

Connection between FD approximations of u' and u''

On a uniform grid the second FD derivative can be written via the first one-sided FD derivatives as:

$$\mathcal{D}_{xx}^{(2)} u_i = \frac{\mathcal{D}_x^+ u_i - \mathcal{D}_x^- u_i}{h} = \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2}$$

FD method in a nutshell

Continuous problem:

$$-\frac{\partial^2 u}{\partial x^2} = f, \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

case 1: $f(x) = -1$

case 2: $f(x) = -x + x^2 - x^3 + x^4 - x^5 + x^6$

case 3: $f(x) = -\sin(\pi x_i)$

FD method: choose $h = 1/N$, define Ω_h on $[0, 1]$, solve

$$-\mathcal{D}_{xx}^{(2)} u_i = f_i, \quad x_i \in \omega_h$$

$$u_0 = 0, \quad u_N = 0$$

case 1: $f_i = -1$

case 2: $f_i = -x_i + x_i^2 - x_i^3 + x_i^4 - x_i^5 + x_i^6$

case 3: $f_i = -\sin(\pi x_i)$

FD method vs analytical solutions

- Since $\mathcal{D}_{xx}^{(2)} u_i$ is an approximation of $u''(x_i)$, the numerical solution of a differential equation obtained by the FD method will be an *approximation* of $u(x_i)$, $i = 1, \dots, N - 1$.
- If we know the analytical solution of the problem, then we can *compute* the error of the FD method (Assignment 2)
- In general we do not know the analytical solution, but we can still *estimate* the error of the FD method (next Lecture)

FDM Step 1: write discrete FD equations

Continuous problem:

$$\begin{aligned}-u'' &= f, \quad x \in \Omega = (a, b); \\ u(a) &= g_0, \quad u(b) = g_1\end{aligned}$$

Discrete FD equations:

$$u_0 = g_0 \quad \text{left BC}$$

$$-\frac{1}{h^2}(u_0 - 2u_1 + u_2) = f_1$$

$$-\frac{1}{h^2}(u_1 - 2u_2 + u_3) = f_2$$

...

$$-\frac{1}{h^2}(u_{N-2} - 2u_{N-1} + u_N) = f_{N-1}$$

$$u_N = g_1 \quad \text{right BC}$$

FDM Step 2: eliminate u_0 and u_N

Boundary values u_0 and u_N are usually eliminated and the BC's are shifted to the right-hand side:

$$\begin{aligned}-\frac{1}{h^2}(-2u_1 + u_2) &= f_1 + \frac{1}{h^2}g_0 \\ -\frac{1}{h^2}(u_1 - 2u_2 + u_3) &= f_2 \\ &\dots \\ -\frac{1}{h^2}(u_{N-2} - 2u_{N-1}) &= f_{N-1} + \frac{1}{h^2}g_1\end{aligned}$$

FDM Step 3: Assemble system matrix and RHS vector

$$\frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ u_{N-1} \end{bmatrix} = \begin{bmatrix} f_1 + g_0/h^2 \\ f_2 \\ \cdot \\ \cdot \\ f_{N-1} + g_1/h^2 \end{bmatrix}$$

The signs correspond to the system matrix of the *negative* 1D Finite-Difference Laplacian. Notice that the system matrix does not depend on g_0 and g_1 .

FDM Step 4: Solve linear algebraic problem

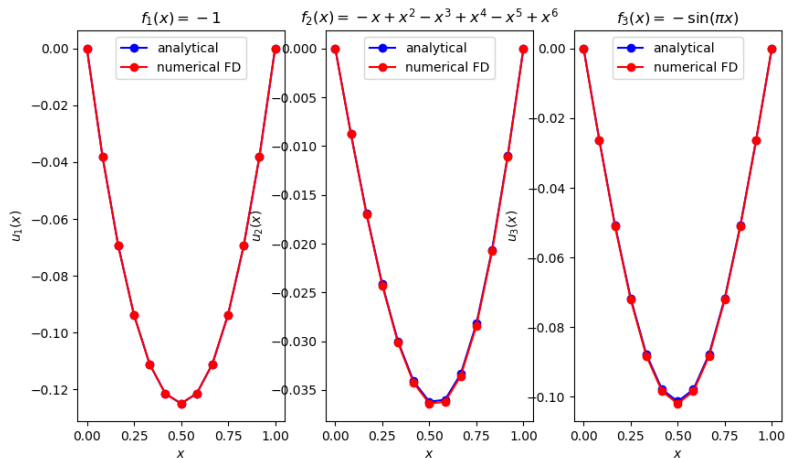
In matrix notation:

$$A\mathbf{u} = \mathbf{f}$$

$$\mathbf{u} = A^{-1}\mathbf{f}$$

The resulting vector $\mathbf{u} \in \mathbb{R}^{N-1}$ contains the FDM approximation of the grid-values of the solution $u(x_i)$ at the inner points of the grid $i = 1, \dots, N - 1$.

Matplotlib figures



Analysis of discretized PDE's

- For the continuous boundary-value problem

$$\begin{aligned}-\Delta u &= f, & \mathbf{x} \in \Omega, \\ u &= g, & \mathbf{x} \in \partial\Omega,\end{aligned}$$

we had the Maximum Principle and established that

- 1 Solution $u(\mathbf{x})$ is unique
 - 2 Sometimes we can tell where the maximum/minimum is and whether $u(\mathbf{x})$ is positive/negative/constant
- After discretization with the Finite-Difference Method we arrived at the linear algebraic problem:

$$L\mathbf{u} = \mathbf{f}$$

- 1 Does it have a unique solution?
- 2 Can we say something about $\mathbf{u}$?
- 3 How big is the error between the numerical u_i and the exact u_i ?

Properties of 1D FD Laplacian

Central-difference 1D FD Laplacian matrix

Consider the matrix L corresponding to the central-difference FD approximation $-\mathcal{D}_{xx}^{(2)}$ of the negative Laplacian operator $-d^2/dx^2$ with Dirichlet BC's:

$$L = \frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & \dots & -1 & 2 \end{bmatrix}$$

What are the main properties?

Positive definiteness

Definition: Matrix $A \in \mathbb{R}^{n \times n}$ is *positive* if $\mathbf{u}^T A \mathbf{u} \geq 0$ and *positive definite* if $\mathbf{u}^T A \mathbf{u} > 0$ for all $\mathbf{u} \in \mathbb{R}^n$, $\mathbf{u} \neq \mathbf{0}$.

Theorem: The 1D FD (negative) Laplacian matrix L is *positive definite*.

Positive definiteness

Proof:

$$\begin{aligned}\mathbf{u}^T L \mathbf{u} &= \frac{1}{h^2} (2u_1^2 - u_1 u_2 - u_1 u_2 + 2u_2^2 - u_2 u_3 + \cdots - u_{N-1} u_{N-2} + 2u_{N-1}^2) \\ &= \frac{1}{h^2} (u_1^2 + u_1^2 - 2u_1 u_2 + u_2^2 + u_2^2 - 2u_2 u_3 + u_3^2 + \\ &\quad \cdots + u_{N-2}^2 - 2u_{N-1} u_{N-2} + u_{N-1}^2 + u_{N-1}^2) \\ &= \frac{1}{h^2} \left(u_1^2 + \sum_{i=1}^{N-2} (u_i - u_{i+1})^2 + u_{N-1}^2 \right) > 0, \quad \text{if } \mathbf{u} \neq \mathbf{0}\end{aligned}$$

Symmetry

Definition: A matrix $A \in \mathbb{R}^{n \times n}$ is *symmetric* if $A^T = A$.

Theorem: The 1D FD (negative) Laplacian matrix L is *symmetric positive definite* (an SPD matrix).

Corollaries:

- 1 There is an eigendecomposition $L = Q\Lambda Q^T$, where $Q^T Q = I$
- 2 All eigenvalues of L are real and positive, also distinct (not a corollary). In particular, the smallest eigenvalue $\lambda_1 > 0$.
- 3 The matrix L is invertible
- 4 There is a unique solution to the problem $Lu = f$

Eigenvalues and eigenvectors

Let the grid step be $h = D/N$, where $D = x_N - x_0$ is the segment length. The eigenvalue problem is

$$L\mathbf{v}_i = \lambda_i\mathbf{v}_i$$

The $N - 1$ eigenvalues of the discrete FD Laplacian matrix $L \in \mathbb{R}^{(N-1) \times (N-1)}$ are given by:

$$\lambda_i = \frac{4}{h^2} \sin^2 \left(\frac{\pi i}{2N} \right); \quad i = 1, 2, \dots, N - 1$$

The corresponding eigenvectors $\mathbf{v}_i$ of L have elements

$$[\mathbf{v}_i]_j = \sin \left(\frac{\pi i j}{N} \right); \quad i, j = 1, 2, \dots, N - 1$$

Correspondence with continuous problem

The eigenvalues and eigenfunctions of the original continuous problem $-v_i'' = \lambda_i v_i$, $v_i(0) = v_i(D) = 0$ are:

$$\tilde{\lambda}_i = \left(\frac{\pi i}{D}\right)^2, \quad v_i(x) = \sin\left(\frac{\pi i x}{D}\right), \quad i = 1, 2, \dots$$

Hence, since $x_j = jh$ and $h = D/N$, the eigenvectors of the FD Laplacian are the exact sampled eigenfunctions of the continuous Laplacian:

$$[\mathbf{v}_i]_j = v_i(x_j)$$

Whereas, the eigenvalues λ_i of the matrix L tend to the eigenvalues $\tilde{\lambda}_i$ of the continuous Laplacian $-d^2/dx^2$ in the limit, as $h \rightarrow 0$:

$$\lim_{h \rightarrow 0} \frac{4}{h^2} \sin^2\left(\frac{\pi i h}{2D}\right) = \left(\frac{\pi i}{D}\right)^2, \quad i = 1, 2, \dots, N-1$$

The smallest eigenvalue

The smallest eigenvalue of L is

$$\lambda_1 = \frac{4}{h^2} \sin^2 \left(\frac{\pi}{2N} \right)$$

The smallest eigenvalue of $-d^2/dx^2$ is

$$\tilde{\lambda}_1 = \left(\frac{\pi}{D} \right)^2$$